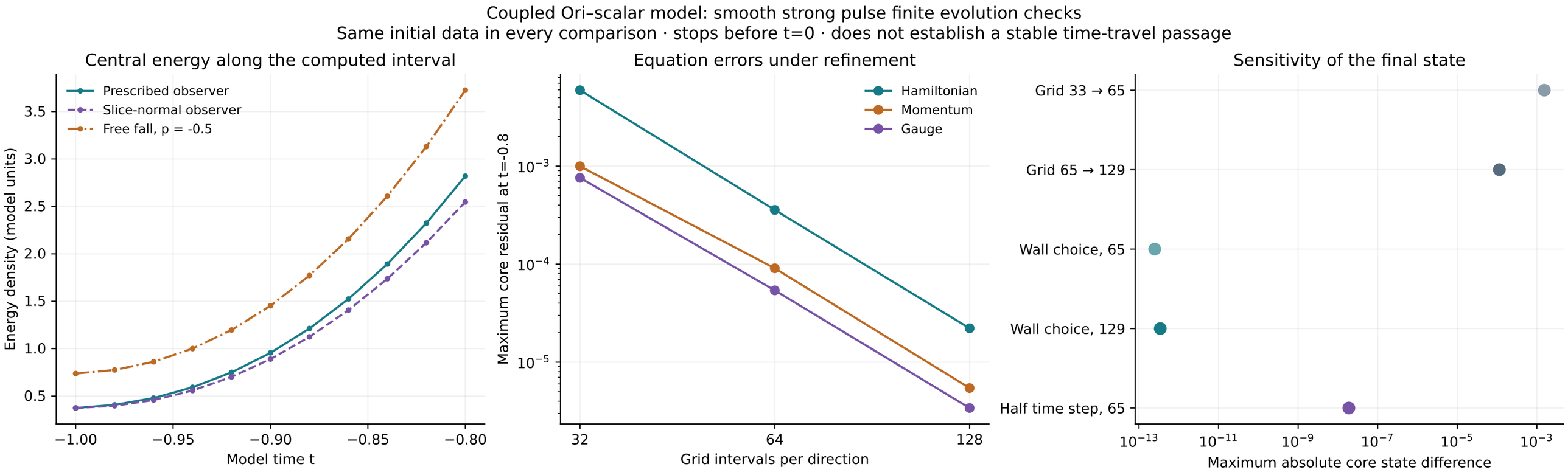
Coupled Ori–scalar model: smooth strong pulse finite evolution checks
Same initial data in every comparison · stops before t=0 · does not establish a stable time-travel passage
Central energy along the computed interval
Prescribed observer
Slice-normal observer
Free fall, p = -0.5
Energy density (model units)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
−1.00
−0.95
−0.90
−0.85
−0.80
Model time t
Equation errors under refinement
Hamiltonian
Momentum
Gauge
Maximum core residual at t=-0.8
10^{-3}
10^{-4}
10^{-5}
32
64
128
Grid intervals per direction
Sensitivity of the final state
Grid 33 → 65
Grid 65 → 129
Wall choice, 65
Wall choice, 129
Half time step, 65
10^{-13}
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
Maximum absolute core state difference